

Sequences & Series



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10} .
 - 2 For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11} .
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Arithmetic series

- 3 Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).
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Geometric sequences

- 4 For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8 .
 - 5 A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.
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Geometric series (finite)

- 6 Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).
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Infinite geometric series

- 7 Find the sum of the infinite geometric series $36 + 12 + 4 + \dots$.
 - 8 Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \dots$.
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Multiple choice — easy

- 9 For the arithmetic sequence with $a_1 = 0$ and common difference $d = 9$, find a_8 .
- a. 63
 - b. 65
 - c. 54
 - d. 72
- 10 For the arithmetic sequence with $a_1 = 17$ and common difference $d = -3$, find a_9 .
- a. -7
 - b. -4
 - c. -5
 - d. -10
- 11 For the arithmetic sequence with $a_1 = -11$ and common difference $d = -5$, find a_6 .
- a. -36
 - b. -34
 - c. -41
 - d. -31
- 12 For the arithmetic sequence with $a_1 = -4$ and common difference $d = -5$, find a_7 .
- a. -34
 - b. -29
 - c. -39
 - d. -32

- 13 For the arithmetic sequence with $a_1 = -6$ and common difference $d = 6$, find a_7 .
- a. 36
 - b. 30
 - c. 32
 - d. 24
- 14 For the arithmetic sequence with $a_1 = -17$ and common difference $d = -6$, find a_{11} .
- a. -83
 - b. -77
 - c. -71
 - d. -75
- 15 For the arithmetic sequence with $a_1 = 15$ and common difference $d = -9$, find a_7 .
- a. -30
 - b. -48
 - c. -37
 - d. -39
- 16 For the arithmetic sequence with $a_1 = 15$ and common difference $d = -2$, find a_{12} .
- a. -7
 - b. -9
 - c. -4
 - d. -5

- 17 For the arithmetic sequence with $a_1 = 1$ and common difference $d = -6$, find a_{10} .
- a. -51
 - b. -47
 - c. -53
 - d. -59
- 18 For the arithmetic sequence with $a_1 = -6$ and common difference $d = -1$, find a_7 .
- a. -13
 - b. -11
 - c. -10
 - d. -12
- 19 For the arithmetic sequence with $a_1 = -15$ and common difference $d = 2$, find a_6 .
- a. -3
 - b. -5
 - c. -7
 - d. -2
- 20 For the arithmetic sequence with $a_1 = -3$ and common difference $d = 9$, find a_7 .
- a. 53
 - b. 42
 - c. 51
 - d. 60

- 21 For the arithmetic sequence with $a_1 = 12$ and common difference $d = -6$, find a_6 .
- a. -12
 - b. -18
 - c. -24
 - d. -16
- 22 For the arithmetic sequence with $a_1 = 4$ and common difference $d = -3$, find a_{13} .
- a. -30
 - b. -32
 - c. -35
 - d. -29
- 23 For the arithmetic sequence with $a_1 = -16$ and common difference $d = -9$, find a_7 .
- a. -68
 - b. -61
 - c. -70
 - d. -79

Multiple choice — medium

- 24 Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.
- a. -94
 - b. -89
 - c. -97
 - d. -93

- 25 Find S_{12} for the arithmetic sequence with $a_1 = 10$, $d = 8$.
- a. 652
 - b. 644
 - c. 648
 - d. 656
- 26 Find S_{14} for the arithmetic sequence with $a_1 = 10$, $d = 6$.
- a. 682
 - b. 686
 - c. 692
 - d. 690
- 27 Find S_4 for the arithmetic sequence with $a_1 = 2$, $d = -7$.
- a. -30
 - b. -41
 - c. -38
 - d. -34
- 28 Find S_4 for the arithmetic sequence with $a_1 = -17$, $d = -5$.
- a. -98
 - b. -102
 - c. -103
 - d. -94

- 29 Find S_{10} for the arithmetic sequence with $a_1 = 3$, $d = -8$.
- a. -326
 - b. -330
 - c. -338
 - d. -334
- 30 Find S_{12} for the arithmetic sequence with $a_1 = 14$, $d = -9$.
- a. -430
 - b. -435
 - c. -422
 - d. -426
- 31 Find S_8 for the arithmetic sequence with $a_1 = 1$, $d = -5$.
- a. -132
 - b. -136
 - c. -128
 - d. -137
- 32 Find S_{10} for the arithmetic sequence with $a_1 = -11$, $d = -6$.
- a. -376
 - b. -386
 - c. -380
 - d. -384

33 Find S_{20} for the arithmetic sequence with $a_1 = -1$, $d = -4$.

a. -780

b. -774

c. -784

d. -776

34 Find S_{10} for the arithmetic sequence with $a_1 = 20$, $d = -7$.

a. -122

b. -115

c. -119

d. -111

35 Find S_6 for the arithmetic sequence with $a_1 = -19$, $d = -1$.

a. -125

b. -133

c. -130

d. -129

36 Find S_8 for the arithmetic sequence with $a_1 = -15$, $d = -4$.

a. -228

b. -226

c. -236

d. -232

37 Find S_{12} for the arithmetic sequence with $a_1 = -13$, $d = -4$.

- a. -424
- b. -420
- c. -414
- d. -416

38 Find S_{16} for the arithmetic sequence with $a_1 = -12$, $d = 3$.

- a. 168
- b. 164
- c. 171
- d. 172

Multiple choice — hard

39 For the geometric sequence with $a_1 = -8$ and common ratio $r = 2$, find a_6 .

- a. 256
- b. -256
- c. -258
- d. -254

40 For the geometric sequence with $a_1 = -9$ and common ratio $r = -3$, find a_4 .

- a. 243
- b. 241
- c. 245
- d. -243

- 41 For the geometric sequence with $a_1 = -2$ and common ratio $r = -3$, find a_6 .
- a. -486
 - b. 488
 - c. 486
 - d. 484
- 42 For the geometric sequence with $a_1 = -6$ and common ratio $r = -3$, find a_3 .
- a. -54
 - b. 54
 - c. -56
 - d. -52
- 43 For the geometric sequence with $a_1 = 9$ and common ratio $r = 3$, find a_3 .
- a. -81
 - b. 79
 - c. 81
 - d. 83
- 44 Find S_5 for the geometric sequence with $a_1 = -3$, $r = 3$.
- a. -361
 - b. -363
 - c. -366
 - d. -365

45 Find S_4 for the geometric sequence with $a_1 = -6$, $r = 3$.

a. -240

b. -238

c. -246

d. -242

46 Find S_7 for the geometric sequence with $a_1 = -5$, $r = 3$.

a. -5467

b. -5465

c. -5470

d. -5463

47 Find S_6 for the geometric sequence with $a_1 = 8$, $r = 3$.

a. 2920

b. 2912

c. 2914

d. 2910

48 Find S_5 for the geometric sequence with $a_1 = -1$, $r = 2$.

a. -29

b. -33

c. -32

d. -31

49 Find the sum of the infinite geometric series $216 - 36 + 6 \dots$.

a. $\frac{1296}{7}$

b. $\frac{1310}{7}$

c. $\frac{1282}{7}$

d. $-\frac{1296}{7}$

50 Find the sum of the infinite geometric series $16 - 8 + 4 \dots$.

a. $\frac{38}{3}$

b. $\frac{32}{3}$

c. $\frac{26}{3}$

d. $-\frac{32}{3}$

51 Find the sum of the infinite geometric series $96 + 24 + 6 \dots$.

a. -128

b. 130

c. 126

d. 128

52 Find the sum of the infinite geometric series $144 - 24 + 4 \dots$.

a. $\frac{878}{7}$

b. $\frac{850}{7}$

c. $\frac{864}{7}$

d. $-\frac{864}{7}$

53 Find the sum of the infinite geometric series $108 - 18 + 3 \dots$.

a. $-\frac{648}{7}$

b. $\frac{634}{7}$

c. $\frac{662}{7}$

d. $\frac{648}{7}$